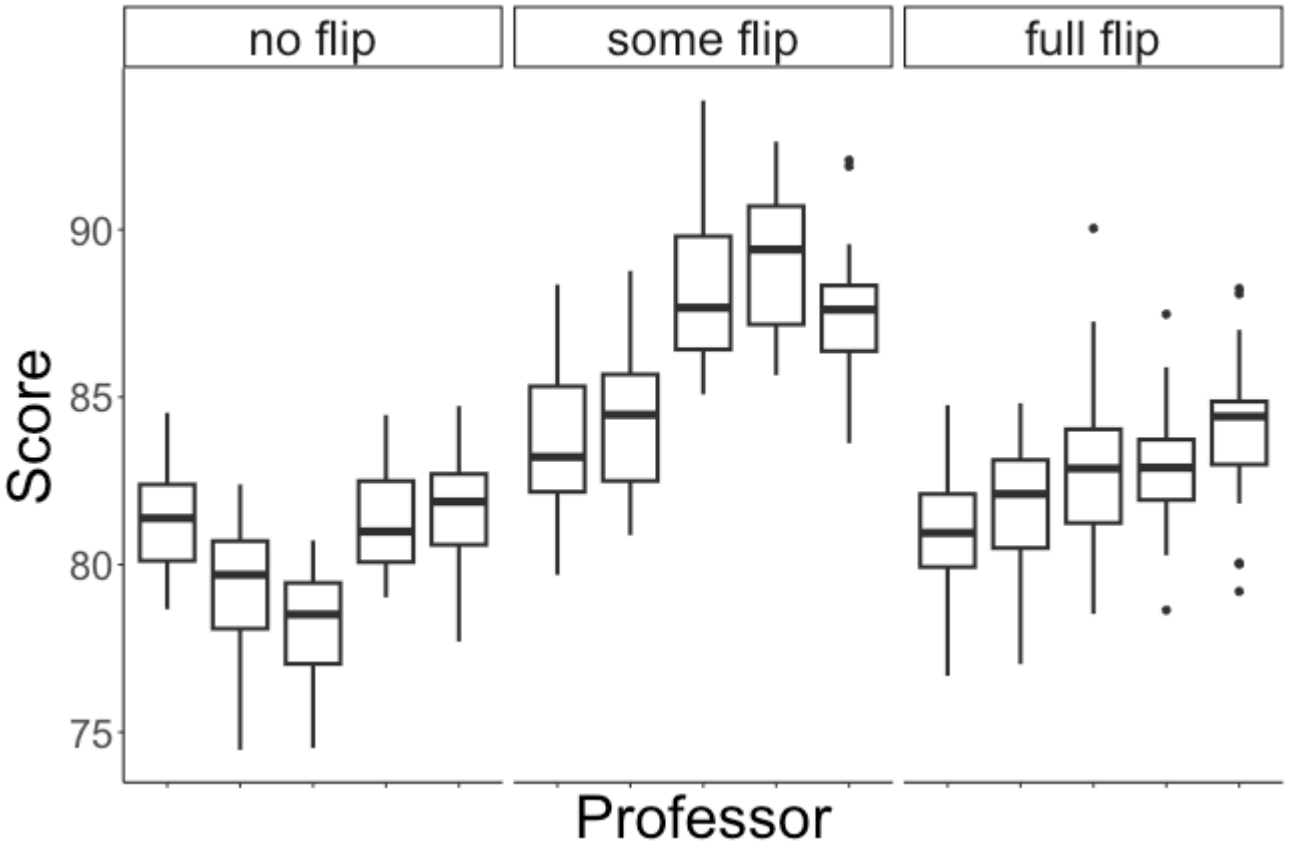
# Beginning linear mixed effects models

# Data: flipped classrooms?

Data set has 375 rows (one per student), and the following variables:

- `professor`: which professor taught the class (1 -- 15)
- `style`: which teaching style the professor used (no flip, some flip, fully flipped)
- `score`: the student's score on the final exam

# Visualizing the data



# Mixed effects model

**Linear mixed effects model:** Let $Score_{ij}$ be the score of student $j$ in class $i$

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \qquad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

# Class activity

Work on the class activity (handout), then we will discuss as a class.

# Class activity

> Why is a mixed effect model useful for this data?

# Class activity

> What is the population model?

# Class activity

> What is the population model?

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

where $Price_{ij}$ is the price of rental $j$ in neighborhood $i$

- $u_i$ is a *random intercept*
- We use subscripts $i$ and $j$ for $Price_{ij}$, $Satisfaction_{ij}$, and $\varepsilon_{ij}$ because they are different for every observation in the data
- We only need a subscript $i$ (neighborhood) for $u_i$, because there is one random intercept per neighborhood

# Class activity

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$u_i \overset{iid}{\sim} N(0, \sigma_u^2)$ $\quad$ $\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$

where $Price_{ij}$ is the price of rental $j$ in neighborhood $i$

> What are the effect of interest, group effect, and individual effect?

# Class activity

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

where $Price_{ij}$ is the price of rental $j$ in neighborhood $i$

> What are the effect of interest, group effect, and individual effect?

- effect of interest: $\beta_1$ (slope for relationship between satisfaction and price)
- group effect: $u_i$ (random effect for neighborhood)
- individual effect: $\varepsilon_{ij}$ (variation between rentals in a neighborhood)

# Fitting mixed effects models

```
library(lme4)
m1 <- lmer(score ~ style + (1|professor),
           data = teaching)
summary(m1)
```

- `lme4` is the R package we will use to fit mixed effects models
- `lmer` is like the `lm` function, but for mixed effects
- `style` is the teaching style (fixed effects)
- `(1|professor)` indicates we have a random intercept (the `1` indicates the intercept) for `professor` (indicated by `|professor`)

# Fitting mixed effects models

```
library(lme4)
```

```
m1 <- lmer(score ~ style + (1|professor),
           data = teaching)
summary(m1)
```

```
Random effects:
 Groups    Name        Variance Std.Dev.
 professor (Intercept) 21.365   4.622
 Residual                4.252   2.062
Number of obs: 375, groups:  professor, 15
```

- $\widehat{\sigma}^2_\varepsilon = 4.25$
- $\widehat{\sigma}^2_u = 21.37$

# Fitting mixed effects models

```
m1 <- lmer(score ~ style + (1|professor),
           data = teaching)
summary(m1)
```

```
Fixed effects:
                Estimate Std. Error t value
(Intercept)       77.657      2.075  37.419
stylesome flip    11.073      2.935   3.773
stylefull flip     2.805      2.935   0.956
```

- $\widehat{\beta}_0 = 77.66$
- $\widehat{\beta}_1 = 11.07$
- $\widehat{\beta}_2 = 2.81$

# Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \quad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \quad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\beta}_0 = 77.66$?

# Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

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- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \quad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\beta}_1 = 11.07$ and $\widehat{\beta}_2 = 2.81$?

# Interpretation

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- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \quad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\beta}_1 = 11.07$ and $\widehat{\beta}_2 = 2.81$?

We expect that, on average, scores in some-flipped classes are 11.07 points higher than for no-flipped classes.

We expect that, on average, scores in fully-flipped classes are 2.81 points higher than for no-flipped classes.

# Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \qquad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \qquad \widehat{\beta}_1 = 11.07, \qquad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \qquad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\sigma}^2_u = 21.37$?

# Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \qquad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \qquad \widehat{\beta}_1 = 11.07, \qquad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \qquad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\sigma}^2_u = 21.37$?

The average score varies from professor to professor by about 4.62 ( $= \sqrt{21.37}$ ) points

# Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \quad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \quad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\sigma}^2_\varepsilon = 4.25$?

# Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

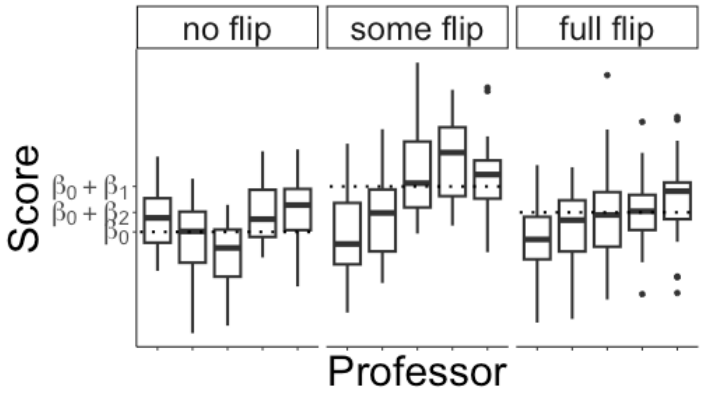
$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \quad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

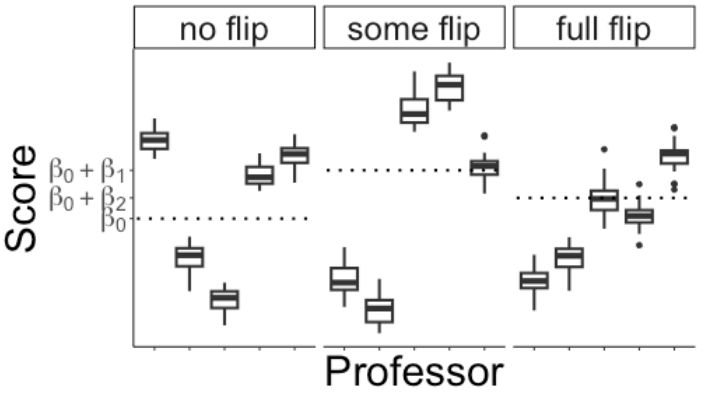
- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \quad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\sigma}^2_\varepsilon = 4.25$?

Within a class, students' scores vary by about 2.06 ( $= \sqrt{4.25}$ ) points

# Intra-class correlation




$\sigma^2_\varepsilon$ is large relative to $\sigma^2_u$

$\sigma^2_\varepsilon$ is small relative to $\sigma^2_u$

- Observations within a group are *more correlated* when $\sigma^2_\varepsilon$ is small relative to $\sigma^2_u$
- **Intra-class correlation:**

$$\rho_{group} = \frac{\sigma^2_u}{\sigma^2_u + \sigma^2_\varepsilon} = \frac{\text{between group variance}}{\text{total variance}}$$

# Intra-class correlation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \quad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \quad \widehat{\sigma}^2_u = 21.37$

$$\hat{\rho}_{group} = \frac{21.37}{21.37 + 4.25} = 0.83$$

So 83% of the variation in student's scores can be explained by differences in average scores from class to class (after accounting for teaching style). That's huge!

# Class activity

Work on the class activity (handout).

# Class activity

Interpret the estimate fixed effect coefficients $\widehat{\beta}_0$ and $\widehat{\beta}_1$.

# Class activity

> Interpret the estimate fixed effect coefficients $\widehat{\beta}_0$ and $\widehat{\beta}_1$.

On average (across neighborhoods), we expect that the price of rental with 0 overall satisfaction is $27.28.

For a fixed neighborhood, an increase of 1 point in overall satisfaction is associated with an increase of $14.81 in rental price.

# Class activity

> Calculate and interpret the estimated intra-class correlation.

# Class activity

> Calculate and interpret the estimated intra-class correlation.

$$\hat{\rho}_{group} = \frac{1048}{1048 + 6762} = 0.134$$

About 13% of the variability in price can be explained by differences in the average price between neighborhoods (after accounting for overall satisfaction).